Lab 6 — Code and Branch Coverage Report

Module under test: `ski_resort_pass.py` — `process_resort_bookings()` and `check_chairlift_anemometer()` (59 statements, 42 branch arcs)

Suite	Tests	Line coverage	Branch coverage	Mutants caught
Starter (`test_ski_resort.py`)	1	47%	44%	—
CodeCoverageTest	17	100% (59/59)	100% (42/42)	14 / 27
BranchCoverageTest	89	100% (59/59)	100% (42/42)	27 / 27

1. How to reproduce

Install the tools:

```
pip install coverage pytest
```

Code (line) coverage:

```
python -m coverage run -m pytest CodeCoverageTest_ski_resort_pass_application.py
python -m coverage report -m
```

Branch coverage:

```
python -m coverage run --branch -m pytest BranchCoverageTest_ski_resort_pass_application.py
python -m coverage report -m
```

2. Starting point

Starter file `test_ski_resort.py` (1 test).

Metric	Result
Line coverage of `ski_resort_pass.py`	47% (31 of 59 statements never run)
Branch coverage of `ski_resort_pass.py`	44% (42 branch arcs, 12 only partly taken)

Never executed: guard clause, inactive groups, season / half-day / unknown durations, child / senior pricing, rejected ages, blackout handling, gear rental, locker fee, AUDIT REVIEW, ZERO SALES, and the whole of `check_chairlift_anemometer`.

3. Code coverage suite

`CodeCoverageTest_ski_resort_pass_application.py`

Metric	Result
Tests	17, all passing
Line coverage of `ski_resort_pass.py`	100% (59 / 59 statements, 0 missed)

Analysis. Line coverage only asks whether a statement ran, so the suite has one focused test per outcome:

- the ValueError guard, and an empty and an inactive booking list;

- each of the three pass durations, plus an unknown one;
- child / senior / adult pricing, and a rejected age;
- blackout override on a full-day and a half-day pass;
- gear rental and the lodge locker;
- all three status codes;
- the three anemometer results (NO DATA, SUSPEND LIFT, SAFE OPERATION).

Expected values were worked out by hand from the pricing rules (for example child full-day = 80 × 0.60 = 48), not copied from the program's output.

4. Branch coverage suite

`BranchCoverageTest_ski_resort_pass_application.py`

Metric	Result
Tests	89 (17 carried over + 72 new, parametrised), all passing
Line coverage of `ski_resort_pass.py`	100% (59 / 59)
Branch coverage of `ski_resort_pass.py`	100% (42 / 42 branch arcs, 0 partial)

What was added. The file is self-contained (it keeps the code-coverage tests) and adds tests so that every decision is taken on both its True and False side:

- **Guard clause** `not isinstance(groups, list) or facility_fee < 0`: each operand tested on its own (None, str, tuple, dict, int as groups; -0.01, -1, -100 as fee) and the fee = 0 boundary.
- **.get() defaults:** missing `is_active`, `duration`, `skiers`, `age`, `gear_rental`, `blackout_override`.
- **Duration if / elif / else:** all arms, plus near-misses such as "FULL DAY", "halfday", None.
- **Age boundaries:** -1 / 0 / 1, 11 / 12, 64 / 65.
- **Blackout override:** half-day break vs other durations, priority over the child / senior discount, interaction with gear rental and the locker fee, and a break voiding only its own group.
- **Status decisions:** exactly 4 vs 5 rejections (> 4), and AUDIT REVIEW taking priority over ZERO SALES.
- **Rounding:** 10.126 → 10.13 and 10.124 → 10.12.
- **Anemometer:** 45 vs 46 mph, a 2-reading streak, streak reset by a calm reading, streak not carried from one lift to the next, empty lifts, None input.

5. Why a second suite was worth writing

coverage.py measures branches as line-to-line jumps, and the line-oriented suite already took every jump, so it also scores 100% branch coverage. A percentage alone therefore cannot tell the two suites apart. To check whether the assertions are actually strong, 27 single-edit bugs ("mutants") were applied one at a time to a scratch copy of the module:

Suite	Mutants caught	Tests
CodeCoverageTest	14 / 27	17

Suite	Mutants caught	Tests
BranchCoverageTest	27 / 27	89

The 13 mutants the line-oriented suite missed were all boundary, default-value or priority bugs:

- `facility_fee < 0` → `<= 0`
- `age < 12` → `<= 12`
- `age >= 65` → `> 65`
- `rejected_passes > 4` → `>= 4`
- the audit check ignoring zero-revenue runs
- `speed > 45` → `>= 45`
- the gust streak carrying over between lifts
- changed defaults for `duration`, `is_active`, `includes_lodge_locker` and `age`
- rounding to 1 dp instead of 2
- `if wind_streams` instead of `if not wind_streams`

These decisions are executed in both directions by the line suite but never at the value where they flip, which is exactly what the branch suite targets.

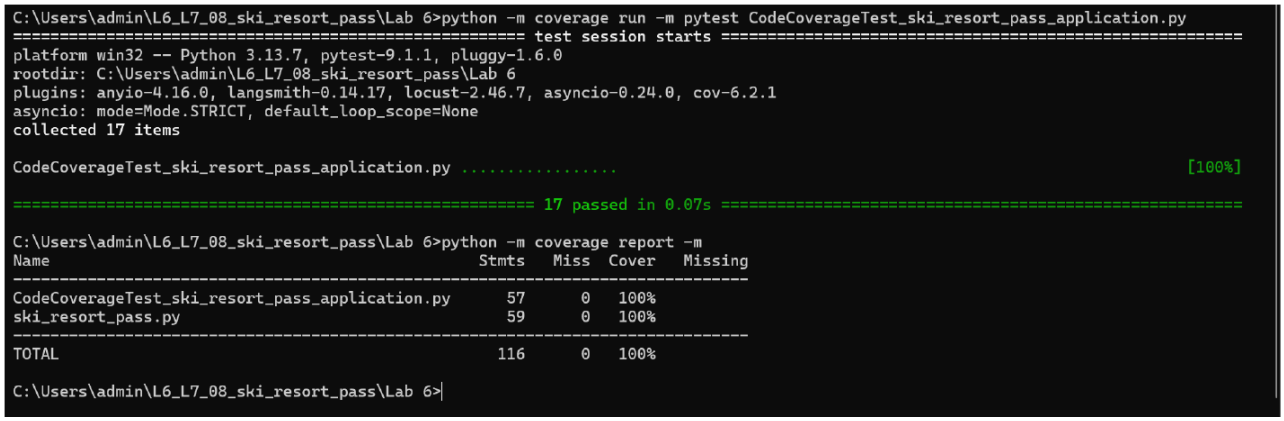
6. Module behaviour worth confirming against the spec

These are pinned by tests as the current behaviour; they may or may not be intended:

1. A blackout-override skier is counted as a rejected pass yet is still billed 50% of the base rate (on non-half-day passes).
2. On a half-day pass a blackout skier voids the group total, but the locker fee is still added afterwards, so a "voided" group can still produce revenue equal to `facility_fee`.
3. An unknown-duration group is skipped before the locker decision, so it never pays the locker fee.

7. Screenshots

Code coverage run (17 passed, 100%):



Branch coverage run (89 passed, 100% branch):

